How do speeds compare?

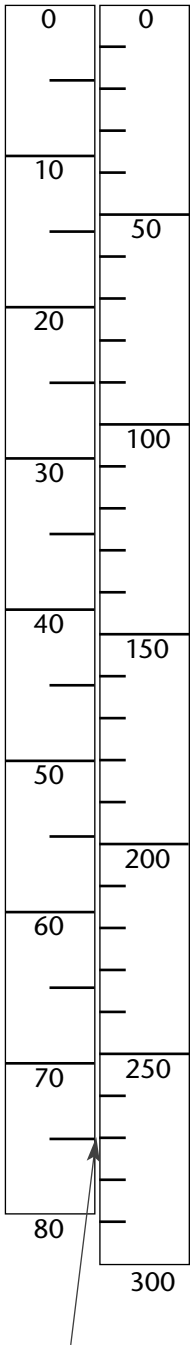
2a

Speed up your calculations by using a calculator.

A champion adult runner covered 100 m in 10 s, which gives his average speed as 10 m/s. (Average speed = 100/10 = 10 m/s) A leopard runs at a speed of 80 km/h for a short time. Who is the faster runner?

The speed converter column and the speed converter graph below relate speeds measured in km/h to the same speeds measured in m/s. This means that they help you to convert between speeds in m/s and km/h.

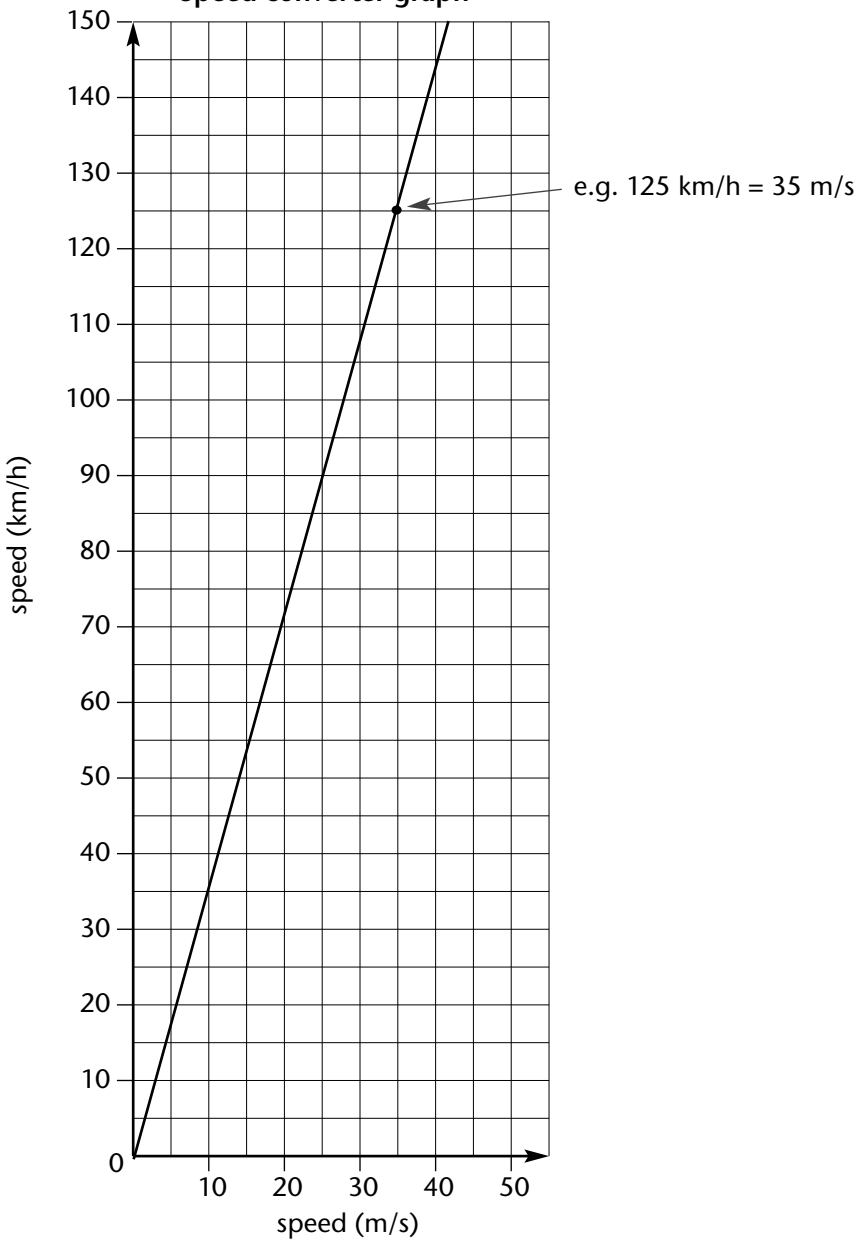
Speed converter column



1 Complete the following table using the speed converter column.

Speed in km/h	Speed in m/s
0 (waiting at the traffic lights)	
	10 (running really, really fast)
80 (a leopard at full speed)	
	1 (walking slowly)
60 (a car driving at town speed limit)	
	70 (speed of a golf ball)

Speed converter graph



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How do speeds compare? 2b

2 Complete the following table using the speed converter graph on the previous page.

Speed in km/h	Speed in m/s
40 (speed limit near a school)	
	6 (running a 400 m race)
110 (speed limit on highways)	
	2 (a fly in full flight)
20 (riding a pushbike)	
	20 (ball speed after a 7 storey drop)

You can convert measurements in one unit to measurements in another unit more accurately by using a calculator. In order to change a speed of 1 metre per second to kilometres per hour, the metre is replaced by 1/1000 km (there are 1000 m in 1 km, so there is 1/1000 km in 1 m) and the second is replaced by 1/3600 of an hour (there are 60 × 60 = 3600 seconds in 1 hour, so there is 1/3600 h in 1 s).

So: 1 m/s = 1 × 1/1000 km /1/3600 h = 1 × 3600/ 1000 km/h = 3.6 km/h

Multiplying the magnitude of a m/s speed by a factor of 3.6 will give the magnitude of the equivalent speed in km/h.

Similarly, to change a speed of 1 kilometre per hour to a speed written in metres per second, parallel substitutions are made:

1 km/h = 1 × 1000 m/3600 s = 1 × 1000/3600 m/s = 1/3.6 m/s = 0.28 m/s

Multiplying the magnitude of a km/h speed by a factor of 0.28 will give the magnitude of the equivalent speed in m/s. Alternatively, you can divide the km/h speed by 3.6 to find the equivalent m/s speed.

3 Calculate equivalent speeds for each speed mentioned in the table. Record the calculations and the speeds calculated in the appropriate cells in the table below.

Speed in km/h	Speed in m/s
1000 (speed of a jet aeroplane)	
	300 (speed of sound)
100 (average speed in quarter mile race)	
	300 000 000 (speed of light)
372 (Formula 1 race track speed record)	
	1500 (speed of a bullet)

4 Calculate the km/h equivalents to the following m/s speeds. Compare your calculated values with those given in the speed converter column on the previous page.

m/s	Calculation	Calculated speed km/h	Converter column km/h
0			
10			
20			
30			
40			
50			
80			

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